

Unit 2: Big Data Infrastructures (Compute/Storage) — **Question Bank**

1 Mark MCQs

Topics covered: Hadoop Ecosystem, Apache Hive, Apache HBase, Apache Cassandra, Apache Iceberg, Apache SeaTunnel, Apache Druid, Apache Hudi, Apache Airflow

50 Questions

(Q1–Q30)

Q1. [1 mark]

What is the primary role of HDFS in the Hadoop ecosystem?

- A)** It schedules containers and allocates CPU/memory across running applications.
- B)** It provides a SQL-like language for querying structured data.
- C)** It stores large files as replicated blocks across a cluster.
- D)** It coordinates scheduled ETL workflows using dependency graphs.

Ans:

The correct answer is C — HDFS (Hadoop Distributed File System) splits large files into blocks and replicates them across DataNodes for durability and scalability.

Q2. [1 mark]

Which Hadoop component is responsible for managing cluster resources and scheduling application containers?

- A)** The HDFS NameNode, which tracks block-to-DataNode mappings.
- B)** YARN, which allocates cluster CPU and memory to applications.
- C)** The Hive Metastore, which stores schema and partition metadata.
- D)** Apache Sqoop, which transfers data between Hadoop and RDBMS.

Ans:

The correct answer is B — YARN acts as the resource management layer, allocating CPU and memory to applications such as MapReduce, Spark, or Hive jobs.

Q3. [1 mark]

In HDFS, how is a large file physically stored across the cluster?

- A)** It is split into fixed-size blocks distributed across DataNodes.
- B)** It is compressed into one archive on a single node.
- C)** It is kept as a single unsplit file on one DataNode.
- D)** It is stored only in the NameNode's local memory.

Ans:

The correct answer is A — HDFS splits files into fixed-size blocks (128 MB by default) and distributes replicated copies of each block across different DataNodes.

Q4. [1 mark]

Which HDFS daemon keeps track of the filesystem namespace and the mapping of blocks to DataNodes?

- A)** The ResourceManager, which tracks running containers.
- B)** The DataNode, which stores metadata locally on each machine.
- C)** The NodeManager, which reports CPU usage per node.
- D)** The NameNode, which maintains the directory tree and block locations.

Ans:

The correct answer is D — the NameNode holds the HDFS metadata, including the directory structure and which DataNodes hold each block's replicas.

Q5. [1 mark]

What kind of interface does Apache Hive provide for querying data stored in Hadoop?

- A)** A drag-and-drop visual query builder with no coding.
- B)** A SQL-like query language called HiveQL.
- C)** A low-level key-value API similar to a NoSQL store.
- D)** A graph traversal language for relationship queries.

Ans:

The correct answer is B — Hive provides HiveQL, a SQL-like language that lets analysts query large datasets stored in HDFS or other storage systems.

Q6. [1 mark]

By default, how does classic Hive execute a HiveQL query behind the scenes?

- A)** It executes the query entirely within the client's memory.
- B)** It compiles the query into one or more MapReduce jobs.
- C)** It sends the query directly to a relational database engine.
- D)** It converts the query into a graph traversal plan.

Ans:

The correct answer is B — traditionally, Hive translates HiveQL statements into one or more MapReduce jobs (though newer versions can also use Tez or Spark).

Q7. [1 mark]

Which Hive component stores table schema information such as column names, types, and partition details?

- A)** ZooKeeper, which only coordinates distributed locks.
- B)** The HMaster, which manages HBase RegionServers.
- C)** The Hive Metastore, a dedicated schema/metadata repository.
- D)** The NameNode, which stores metadata for HDFS blocks.

Ans:

The correct answer is C — the Hive Metastore is a separate service/database that stores schema, partition, and table-location metadata used by Hive.

Q8. [1 mark]

Apache HBase's architecture is closely modeled after which system originally described by Google?

- A)** Bigtable, Google's distributed column-oriented storage system.
- B)** Dremel, Google's interactive query engine.
- C)** Spanner, Google's globally distributed SQL database.
- D)** Google File System, Google's distributed storage paper.

Ans:

The correct answer is A — HBase is an open-source system inspired by Google's Bigtable design, offering distributed, column-oriented storage.

Q9. [1 mark]

How would you best classify Apache HBase as a database system?

- A)** A relational database enforcing strict row-level schemas.
- B)** A column-oriented, wide-column NoSQL database.
- C)** A pure in-memory cache with no persistence layer.
- D)** A document-oriented database storing JSON documents.

Ans:

The correct answer is B — HBase stores data using column families in a wide-column NoSQL model, optimized for fast random reads and writes.

Q10. [1 mark]

Which underlying storage system does HBase typically rely on for durable, distributed data storage?

- A)** An external relational database exclusively.
- B)** In-memory storage with no persistence layer.
- C)** HDFS, which provides distributed and replicated storage.
- D)** A local filesystem on a single dedicated machine.

Ans:

The correct answer is C — HBase is usually deployed on top of HDFS, which gives it distributed, replicated, and fault-tolerant storage for its data files.

Q11. [1 mark]

What consistency model does Apache Cassandra follow by design?

- A)** No consistency guarantees are ever provided at all.
- B)** Consistency achievable only through a single master node.
- C)** Strict serializability enforced on every single operation.
- D)** Tunable consistency, chosen per operation by the client.

Ans:

The correct answer is D — Cassandra offers tunable consistency, letting applications choose a consistency level per read or write to balance consistency and availability.

Q12. [1 mark]

How would you describe the overall architecture of a Cassandra cluster?

- A)** A single powerful node handling all reads and writes.
- B)** A peer-to-peer, masterless ring where all nodes are equal.
- C)** A tree-structured hierarchy of workers and managers.
- D)** A strict master-slave hierarchy with one coordinator node.

Ans:

The correct answer is B — Cassandra uses a masterless, peer-to-peer architecture in which all nodes are functionally equal and arranged in a ring.

Q13. [1 mark]

Which query language is used to interact with data stored in Apache Cassandra?

- A)** PL/SQL, Oracle's procedural extension to SQL.
- B)** Cypher, a language designed for graph databases.
- C)** HiveQL, borrowed directly from the Hadoop ecosystem.
- D)** CQL, Cassandra's own SQL-like query language.

Ans:

The correct answer is D — Cassandra provides CQL, a query language with syntax similar to SQL, tailored to Cassandra's data model.

Q14. [1 mark]

How is Apache Iceberg best described within the big data ecosystem?

- A)** A table format adding ACID transactions and schema evolution to data lakes.
- B)** A message broker used for real-time event streaming.
- C)** A graph database optimized for relationship queries.
- D)** A resource scheduler that allocates cluster CPU and memory.

Ans:

The correct answer is A — Iceberg is a table format designed to bring reliable, SQL-like ACID transactions and safe schema evolution to data lakes.

Q15. [1 mark]

Which capability does Apache Iceberg add on top of plain files stored in a data lake?

- A)** Automatic conversion of all data into an in-memory cache.
- B)** Reliable schema evolution and ACID-compliant transactions.
- C)** Built-in graph traversal algorithms for relationships.
- D)** Real-time video encoding and transcoding support.

Ans:

The correct answer is B — Iceberg was specifically designed to solve problems like unsafe schema changes and the lack of transactional guarantees in raw data lakes.

Q16. [1 mark]

What is the primary purpose of Apache SeaTunnel in a big data architecture?

- A)** Distributed, high-performance data integration between many sources.
- B)** Rotating and archiving application log files.
- C)** Training and serving machine learning models at scale.
- D)** Managing container orchestration for microservices.

Ans:

The correct answer is A — SeaTunnel is built for synchronizing large volumes of data across many different, heterogeneous data sources and sinks.

Q17. [1 mark]

How many different data sources can Apache SeaTunnel typically synchronize data between?

- A)** Only between Kafka topics within a single cluster.
- B)** Only between flat CSV files on the same machine.
- C)** Only between exactly two fixed databases at a time.
- D)** Hundreds of different, heterogeneous data sources.

Ans:

The correct answer is D — SeaTunnel is designed to synchronize huge amounts of data across hundreds of different data sources through its connector framework.

Q18. [1 mark]

Which description best fits Apache Druid's role in a big data stack?

- A)** A workflow scheduler used to sequence dependent tasks.
- B)** A high-performance analytics database for real-time and historical data.
- C)** A purely batch-oriented ETL tool with no query capability.
- D)** A distributed version control system for datasets.

Ans:

The correct answer is B — Druid is a distributed analytics database engineered to serve fast queries over both streaming and previously ingested data.

Q19. [1 mark]

What type of use case is Apache Druid particularly well optimized for?

- A)** Fast, interactive OLAP-style aggregation queries for dashboards.
- B)** Complex multi-hop graph traversal and pathfinding queries.
- C)** High-resolution video transcoding pipelines.
- D)** Long-term, rarely-accessed cold archival storage of raw files.

Ans:

The correct answer is A — Druid is optimized for sub-second aggregation queries over large event/time-series datasets, popular for real-time dashboards.

Q20. [1 mark]

What core capability does Apache Hudi add to data stored in a data lake?

- A)** Real-time record-level updates and incremental processing.
- B)** Container-level orchestration of compute resources.
- C)** Message queuing between distributed microservices.
- D)** Static, write-once storage with no update support.

Ans:

The correct answer is A — Hudi brings the ability to perform real-time upserts and deletes on data lake tables, plus incremental data processing.

Q21. [1 mark]

In the name "Hadoop Upserts Deletes and Incrementals" (Hudi), what does "upsert" refer to?

- A)** A method for merging two unrelated database schemas.
- B)** The process of uploading files to cloud storage.
- C)** A combined update-or-insert operation on a record.
- D)** A user-authentication and login mechanism.

Ans:

The correct answer is C — an upsert is a combined update-or-insert operation: if a record already exists it is updated, and if not, it is inserted.

Q22. [1 mark]

What is the main function of Apache Airflow within a big data pipeline?

- A)** Orchestrating and scheduling complex ETL workflows.
- B)** Providing distributed, replicated block storage for large files.
- C)** Performing real-time streaming analytics on event data.
- D)** Storing data using column families for random access.

Ans:

The correct answer is A — Airflow is a platform for programmatically authoring, scheduling, and monitoring workflows, ideal for coordinating ETL pipelines.

Q23. [1 mark]

How is a workflow typically represented within Apache Airflow?

- A)** As a Directed Acyclic Graph of tasks and dependencies.
- B)** As a binary search tree of scheduled jobs.
- C)** As a simple linear list of commands with no branching.
- D)** As a flat hash map of unrelated, independent tasks.

Ans:

The correct answer is A — Airflow workflows are defined as DAGs, where nodes represent tasks and directed edges define the order in which they must run.

Q24. [1 mark]

Which of these options is NOT actually part of the Hadoop ecosystem tools discussed for compute and storage?

- A)** MapReduce, used as a distributed batch processing model.
- B)** A desktop image-editing application unrelated to data processing.
- C)** YARN, used for cluster resource management.
- D)** HDFS, used for distributed file storage.

Ans:

The correct answer is B — a desktop image-editing application has nothing to do with Hadoop's distributed compute or storage stack, unlike HDFS, YARN, and MapReduce.

Q25. [1 mark]

How does HBase organize related data within a single row?

- A)** By relying entirely on foreign-key relationships between tables.
- B)** By using stored procedures to compute column values.
- C)** By grouping related columns into column families stored together.
- D)** By storing every column in a completely separate table.

Ans:

The correct answer is C — HBase groups related columns into column families, which are physically stored together, allowing efficient access to related data.

Q26. [1 mark]

How does Cassandra decide which nodes in the cluster will store a particular row of data?

- A)** By hashing the partition key to a position on a token ring.
- B)** By sending all data to one designated coordinator node.
- C)** By using a single central lock manager for every write.
- D)** By relying on a human administrator to assign rows manually.

Ans:

The correct answer is A — Cassandra applies consistent hashing on the partition key to determine which node(s) on the ring are responsible for storing a given row.

Q27. [1 mark]

In the context of tools like Iceberg and Hudi, what best describes a "data lake"?

- A)** A single small CSV file kept on one analyst's laptop.
- B)** A centralized repository storing large volumes of raw data.
- C)** A purely in-memory cache that loses data on restart.
- D)** A traditional row-based OLTP database used for transactions.

Ans:

The correct answer is B — a data lake is a large-scale repository that stores raw data in its native format, later structured and queried for analytics.

Q28. [1 mark]

How are task dependencies typically expressed between two tasks in an Airflow DAG?

- A)** Using HDFS symbolic links between output directories.
- B)** Using Cassandra consistency tokens shared between tasks.
- C)** Using dependency operators such as the `>>` symbol.
- D)** Using SQL join statements written inside the DAG file.

Ans:

The correct answer is C — Airflow lets developers define task order using dependency-setting syntax such as `task_a >> task_b`.

Q29. [1 mark]

Which of these systems would you choose for fast, low-latency random read/write access to individual records at scale?

- A)** Apache SeaTunnel, designed for batch data synchronization.
- B)** A static CSV file stored on a single machine.
- C)** Apache HBase, built for real-time random access to records.
- D)** Apache Airflow, designed purely for scheduling workflows.

Ans:

The correct answer is C — HBase is specifically built for fast random reads and writes on individual records, unlike orchestration or integration tools.

Q30. [1 mark]

What do the ACID transaction guarantees provided by Apache Iceberg primarily ensure?

- A)** End-to-end encryption of all network traffic.
- B)** Automatic generation of dashboards and visualizations.
- C)** Faster CPU clock speeds on the machines running queries.
- D)** Reliable and consistent changes to table data.

Ans:

The correct answer is D — ACID guarantees ensure that changes made to an Iceberg table are applied reliably and consistently, even with concurrent writers.

(Q31–Q43)

Q31. [1 mark]

What is the default block replication factor used by HDFS, and what is its main purpose?

- A)** A replication factor of 3, balancing fault tolerance and cost.
- B)** A replication factor of 2, balancing cost and safety.
- C)** A replication factor of 1, purely to save on storage costs.
- D)** A replication factor of 5, to maximize redundancy.

Ans:

The correct answer is A — HDFS defaults to a replication factor of 3, providing good fault tolerance against node failures while keeping storage overhead reasonable.

Q32. [Numerical; 1 mark]

A file of size 500 MB is stored in HDFS with the default block size of 128 MB and replication factor 3. Approximately how much total physical storage will this consume, and into how many blocks is the file split?

- A) About 1500 MB total storage, split into 4 blocks.
- B) About 1000 MB total storage, split into 4 blocks.
- C) About 500 MB total storage, split into 3 blocks.
- D) About 1500 MB total storage, split into 5 blocks.

Ans:

The correct answer is A — 500 MB ÷ 128 MB rounds up to 4 blocks (128+128+128+116 MB), and with a replication factor of 3, total storage is approximately $500 \times 3 = 1500$ MB.

Q33. [1 mark]

Besides classic MapReduce, which execution engine is commonly used by Hive to significantly speed up query execution?

- A) Apache Oozie, used only for workflow scheduling.
- B) Apache Flume, used for log collection.
- C) Apache Tez, which models queries as a DAG of tasks.
- D) Apache Kafka, used purely for message streaming.

Ans:

The correct answer is C — Hive can use Apache Tez (or Spark) as an alternative execution engine, reducing disk I/O overhead compared to classic MapReduce.

Q34. [1 mark]

Which HBase component is primarily responsible for assigning regions to RegionServers and coordinating load balancing?

- A) The HMaster, which coordinates RegionServers directly.
- B) The MapReduce JobTracker, inherited from classic Hadoop.
- C) ZooKeeper alone, without any other component involved.
- D) The HDFS NameNode, which assigns regions directly.

Ans:

The correct answer is A — the HMaster coordinates RegionServers, handling tasks such as region assignment, load balancing, and schema changes.

Q35. [1 mark]

How does a client typically control Cassandra's tunable consistency for a specific read or write operation?

- A)** By adjusting the size of the partition key used.
- B)** By setting a different replication factor per query.
- C)** By choosing a specific compaction strategy at query time.
- D)** By specifying a Consistency Level such as ONE or QUORUM.

Ans:

The correct answer is D — clients specify a Consistency Level (e.g., ONE, QUORUM, ALL) per query, determining how many replicas must respond for success.

Q36. [1 mark]

How does Apache Iceberg's "hidden partitioning" feature primarily benefit users writing queries?

- A)** It encrypts all partition values from unauthorized users.
- B)** It automatically compresses every data file regardless of partitioning.
- C)** It removes the concept of partitioning from the table entirely.
- D)** It lets users query by logical columns without knowing the physical layout.

Ans:

The correct answer is D — hidden partitioning lets Iceberg manage partition transforms internally, so analysts can query using natural columns.

Q37. [1 mark]

What is a key architectural advantage of Apache SeaTunnel compared to writing many custom point-to-point ETL scripts?

- A)** It eliminates the need for any data sources or sinks.
- B)** It only supports batch processing and cannot handle streams.
- C)** It offers a pluggable connector framework for many-to-many sync.
- D)** It fully replaces the need for any workflow orchestration tool.

Ans:

The correct answer is C — SeaTunnel's pluggable Source-Transform-Sink connector architecture enables scalable, configuration-driven data synchronization.

Q38. [1 mark]

In Apache Druid, what are the immutable, columnar chunks of indexed data called?

- A)** Blocks, borrowed directly from HDFS terminology.
- B)** Shards, a term used exclusively in Cassandra.
- C)** Pages, a term used in traditional relational databases.
- D)** Segments, Druid's core unit of storage and query optimization.

Ans:

The correct answer is D — Druid stores ingested data as immutable, columnar "segments," optimized for compression and fast analytical scanning.

Q39. [1 mark]

Which Druid component is primarily responsible for running real-time ingestion tasks before data is converted into segments?

- A)** MiddleManager nodes, which run real-time ingestion tasks.
- B)** Historical nodes, which only serve already-finalized segments.
- C)** Broker nodes, which only route and merge query results.
- D)** Coordinator nodes, which only manage segment balancing.

Ans:

The correct answer is A — MiddleManager nodes execute ingestion tasks, including real-time streaming ingestion, before data becomes a segment.

Q40. [1 mark]

Which two table storage types does Apache Hudi support, and how do they primarily differ?

- A)** In-memory and on-disk, differing in durability only.
- B)** Copy-on-Write and Merge-on-Read, differing in merge timing.
- C)** Append-only and overwrite-only, with no partial updates.
- D)** Row-store and column-store, differing in file format only.

Ans:

The correct answer is B — Hudi supports Copy-on-Write, which merges updates at write time, and Merge-on-Read, which defers merging until read time.

Q41. [1 mark]

In Apache Airflow, what is the role of an "Operator" such as a BashOperator or PythonOperator?

- A)** It only determines which server the webserver runs on.
- B)** It is used exclusively to store credentials for external systems.
- C)** It defines the reusable unit of work a task performs.
- D)** It manages the metadata database used by the scheduler.

Ans:

The correct answer is C — Operators define what a task actually does, such as running a bash command or executing Python code.

Q42. [1 mark]

Which Airflow component determines how and where tasks are actually executed, such as via local processes or distributed Celery workers?

- A)** The Scheduler, which only decides when tasks should run.
- B)** The Executor, which governs how and where tasks run.
- C)** The Metadata Database, which only stores task state.
- D)** The Webserver, which only provides the user interface.

Ans:

The correct answer is B — the Executor (e.g., LocalExecutor, CeleryExecutor) governs how and where tasks are actually executed once ready to run.

Q43. [1 mark]

Which statement correctly describes Cassandra's write path when a new write request arrives at a coordinator node?

- A)** The write goes directly to an SSTable, bypassing memory.
- B)** The write skips replication unless explicitly requested.
- C)** The write always requires a single master node's approval.
- D)** The write goes to a commit log and memtable, then flushes to disk.

Ans:

The correct answer is D — Cassandra first appends the write to a commit log for durability and applies it to an in-memory memtable, later flushed as an SSTable.

(Q44–Q50)

Q44. [1 mark]

How does HDFS High Availability (HA) typically avoid a single point of failure at the NameNode?

- A)** By letting YARN's ResourceManager take over NameNode duties.
- B)** By using Active/Standby NameNodes synchronized via JournalNodes.
- C)** By running multiple DataNodes only, without NameNode changes.
- D)** By replicating all NameNode metadata directly into HBase tables.

Ans:

The correct answer is B — HDFS HA uses an Active NameNode and one or more Standby NameNodes, synchronized through JournalNodes using the Quorum Journal Manager.

Q45. [1 mark]

Under what condition does HBase automatically perform a "region split"?

- A)** Whenever the HMaster process is manually restarted.
- B)** Whenever any RegionServer in the cluster crashes.
- C)** When a region grows beyond a configured size threshold.
- D)** Whenever a column family is deleted from the schema.

Ans:

The correct answer is C — HBase automatically splits a region into two once it exceeds a configured size threshold, keeping load balanced across RegionServers.

Q46. [1 mark]

Which Cassandra mechanism allows conflicting writes to the same cell across replicas to be resolved without coordination between nodes?

- A)** A Last-Write-Wins strategy based on write timestamps.
- B)** A centralized distributed lock blocking concurrent writes.
- C)** A manual conflict-resolution interface requiring admin input.
- D)** A strict two-phase commit executed across all replicas.

Ans:

The correct answer is A — Cassandra resolves conflicting writes using a Last-Write-Wins approach based on timestamps, avoiding coordination overhead.

Q47. [1 mark]

How does Apache Iceberg's metadata layer, consisting of manifest lists and manifest files, help query engines during planning?

- A)** It encrypts data files while being transferred over the network.
- B)** It lets engines prune irrelevant files using stored statistics.
- C)** It compresses the actual data files stored on disk.
- D)** It completely removes the need for a catalog service.

Ans:

The correct answer is B — manifest lists/files track data files with partition and column statistics, letting engines prune irrelevant files without costly directory listings.

Q48. [1 mark]

Which Druid node type receives incoming queries, routes them to Historical and real-time nodes, and merges the results?

- A)** The Coordinator, which only manages segment balancing.
- B)** The Broker, which routes queries and merges results.
- C)** The Overlord, which only manages ingestion task assignment.
- D)** The Router, used only for basic HTTP request forwarding.

Ans:

The correct answer is B — Broker nodes receive client queries, forward them to relevant Historical and real-time nodes, and merge the partial results.

Q49. [1 mark]

What trade-off do Merge-on-Read (MOR) tables in Apache Hudi make between write and read performance?

- A)** No trade-off exists between reads and writes at all.
- B)** Faster reads at the cost of disabling update support entirely.
- C)** Faster writes at the cost of higher read latency until compaction.
- D)** Higher storage cost only, with no impact on latency.

Ans:

The correct answer is C — MOR tables write changes quickly as small delta logs, but reads must merge these with base files (until compaction), increasing read latency.

Q50. [1 mark]

What is the primary purpose of "XComs" in Apache Airflow?

- A)** Passing small pieces of data between tasks in a DAG run.
- B)** Scheduling DAGs to run on a fixed cron-like interval.
- C)** Storing large datasets persistently as the primary data store.
- D)** Authenticating users who want to log into the webserver.

Ans:

The correct answer is A — XComs ("cross-communications") let tasks exchange small amounts of data, such as a value one task computes for a downstream task.

Answer Key

Section A (Q1-Q50): 1-C, 2-B, 3-A, 4-D, 5-B, 6-B, 7-C, 8-A, 9-B, 10-C,
11-D, 12-B, 13-D, 14-A, 15-B, 16-A, 17-D, 18-B, 19-A, 20-A, 21-C, 22-A,
23-A, 24-B, 25-C, 26-A, 27-B, 28-C, 29-C, 30-D, 31-A, 32-A, 33-C, 34-A,
35-D, 36-D, 37-C, 38-D, 39-A, 40-B, 41-C, 42-B, 43-D, 44-B, 45-C, 46-A,
47-B, 48-B, 49-C, 50-A